

Transformation of the equation by changing the independent variable

Consider,

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R$$

Now x is changed to z . Then the equation becomes,

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1$$

where;

$$P_1 = \frac{\frac{d^2 z}{dx^2} + P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} \quad \& \quad \left(\frac{dz}{dx}\right)^2 = Q$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2}$$

$$R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2}$$

Questions: Solve

1. $y'' + \frac{2}{x} y' + \frac{a^2}{x^4} y = 0$

Solⁿ:

The given differential eqⁿ. is,

$$y'' + \frac{2}{x} y' + \frac{a^2}{x^4} y = 0 \quad \text{--- (1)}$$

which is a second order differential equation.

Comparing the given eqⁿ. with

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = R$$

$$\Rightarrow P = \frac{2}{x}, \quad Q = \frac{a^2}{x^4} \quad \& \quad R = 0$$

Now, x is changed to z , such that,

$$\left(\frac{dz}{dx} \right)^2 = \frac{a^2}{x^4}$$

$$\Rightarrow \frac{dz}{dx} = \frac{a}{x^2}$$

$$\Rightarrow z = \frac{-a}{x} \quad \text{[After Integration]}$$

~~Now~~ Now, the eqⁿ. becomes

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1 \quad \text{--- (2)}$$

where,

$$P_1 = \frac{d^2 z}{dx^2} + \frac{P dz}{dx} = \frac{-2a}{x^3} + \frac{2}{x} \times \frac{a}{x^2} = 0$$
$$\left(\frac{dz}{dx}\right)^2 = \frac{a^2}{x^4}$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{a^2/x^4}{a^2/x^4} = 1$$

$$R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} = 0$$

$\therefore$ Eqⁿ. (2) becomes,

$$\frac{d^2 y}{dz^2} + y = 0$$
$$\Rightarrow (D^2 + 1)y = 0$$

The A.E. is,

$$m^2 = -1 \Rightarrow m = \pm i$$

$$C.F. = C_1 \cos z + C_2 \sin z$$

$$\therefore y = C_1 \cos z + C_2 \sin z = C_1 \cos \frac{a}{x} - C_2 \sin \frac{a}{x}$$

Ans

$$2. \quad x \frac{d^2 y}{dx^2} - \frac{dy}{dx} - 4x^3 y = 8x^3 \sin x^2$$

$$\Rightarrow \frac{d^2 y}{dx^2} - \frac{1}{x} - 4x^2 y = 8x^2 \sin x^2$$

Solⁿ:

Comparing the given eqⁿ with,

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Qy = R$$

we get,

$$P = -\frac{1}{x}, \quad Q = -4x^2 \quad \& \quad R = 8x^2 \sin x^2$$

Now,

x is changed to z , such that,

$$\left(\frac{dz}{dx} \right)^2 = 4x^2$$

$$\Rightarrow \frac{dz}{dx} = 2x$$

$$\Rightarrow z = x^2$$

Now, we have eqⁿ,

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1$$

$$P_1 = \frac{d^2 z}{dx^2} + \frac{P \frac{dz}{dx}}{\left(\frac{dz}{dx} \right)^2} = 2 + \frac{\left(-\frac{1}{x} \times 2x \right)}{4x^2} = 0$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx} \right)^2} = \frac{-4x^2}{4x^2} = -1$$

$$R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} = \frac{8x^2 \cdot \sin x^2}{4x^2} = 2 \sin x^2$$

$\therefore$ eqⁿ. becomes,

$$\frac{d^2 y}{dz^2} - y = 2 \sin(z)$$

$$\Rightarrow (D^2 - 1)y = 2 \sin(z)$$

The A.E. is,

$$m^2 - 1 = 0$$

$$\Rightarrow m^2 = 1$$

$$\Rightarrow m = \pm 1$$

$$C.F. = C_1 e^z + C_2 e^{-z}$$

$$P.I. = \frac{1}{(D^2 - 1)} \cdot 2 \sin(z)$$

$$= 2 \cdot \frac{1}{-1^2 - 1} \cdot \sin z$$

$$= \frac{2}{-2} \cdot \sin z$$

$$= -\sin z$$

$$\therefore y = C.F. + P.I.$$

$$= C_1 e^z + C_2 e^{-z} - \sin z$$

$$= C_1 e^{x^2} + C_2 e^{-x^2} - \sin x^2 \quad \underline{\text{Ans}}$$

$$3. (1+x)^2 \frac{d^2 y}{dx^2} + (1+x) \frac{dy}{dx} + y = 4 \cos \log(1+x)$$

Solⁿ:

The eqⁿ. can be written as,

$$\frac{d^2 y}{dx^2} + \frac{1}{(1+x)} \frac{dy}{dx} + \frac{1}{(1+x)^2} y = \frac{4 \cos \log(1+x)}{(1+x)^2}$$

Comparing above eqⁿ. with,

$$\frac{d^2 y}{dx^2} + P \frac{dy}{dx} + Q y = R$$

we get,

$$P = \frac{1}{1+x}, \quad Q = \frac{1}{(1+x)^2} \quad \& \quad R = \frac{4 \cos \log(1+x)}{(1+x)^2}$$

Now, x is changed to z ,

$$\left(\frac{dz}{dx} \right)^2 = \frac{1}{(1+x)^2}$$

$$\Rightarrow \frac{dz}{dx} = \frac{1}{1+x} \Rightarrow \frac{d^2 z}{dx^2} = \frac{-1}{(1+x)^2}$$

$$\Rightarrow z = \log(1+x)$$

when x is changed to z , we have the eqⁿ,

$$\frac{d^2 y}{dz^2} + P_1 \frac{dy}{dz} + Q_1 y = R_1$$

$$P_1 = \frac{d^2 z}{dx^2} + P \frac{dz}{dx} = 0$$

$$\left(\frac{dz}{dx} \right)^2$$

$$Q_1 = \frac{0}{\left(\frac{dz}{dx}\right)^2} = 1$$

$$R = \frac{R}{\left(\frac{dz}{dx}\right)^2} = 4 \cos \log(1+x) = 4 \cos z$$

So, eqⁿ. becomes,

$$\frac{d^2 y}{dz^2} + y = \cancel{4 \cos \log(1+x)} = 4 \cos z$$

$$\Rightarrow (D^2 + 1)y = 4 \cos z$$

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$$C.F. = C_1 \cos z + C_2 \sin z$$

$$P.I. = \frac{1}{(D^2 + 1)} \cdot 4 \cos z$$

$$= \frac{4}{-1^2 + 1} \cos z$$

Case of failure

$$\begin{aligned}
 P.I. &= 4z \cdot \frac{1}{2D} \cdot \cos z \\
 &= \frac{4z}{2} \int \cos z \, dz \\
 &= 2z \cdot \sin z
 \end{aligned}$$

$$\therefore y = C.F. + P.I.$$

$$\Rightarrow y = C_1 \cos z + C_2 \sin z + 2z \sin z$$

$$\Rightarrow y = C_1 \cos \log(1+x) + C_2 \sin \log(1+x) + 2 \log(1+x) \sin \log(1+x).$$

$$4. \cos x y'' + y' \sin x - 2y \cos^3 x = 2 \cos^5 x$$

Solⁿ:

The given eqⁿ. can be written as,

$$y'' + \tan x y' - 2 \cos^2 x \cdot y = 2 \cos^4 x$$

Comparing above eqⁿ. with $y'' + P y' + Q y = R$, we get,

$$P = \tan x, Q = -2 \cos^2 x \text{ \& } R = 2 \cos^4 x$$

Now, x is changed to z ,

$$\left(\frac{dz}{dx} \right)^2 = \cos^2 x$$

$$\Rightarrow \frac{dz}{dx} = \cos x \Rightarrow \frac{d^2 z}{dx^2} = -\sin x$$

$$\Rightarrow z = \sin x$$

we can write eqn.,

$$y'' + P_1 y + Q_1 y' = R_1$$

$$P_1 = \frac{\frac{d^2 z}{dx^2}}{\left(\frac{dz}{dx}\right)^2} + \frac{P \frac{dz}{dx}}{\left(\frac{dz}{dx}\right)^2} = \frac{-\sin x + \tan x \cdot \cos x}{\cos^2 x} = 0$$

$$Q_1 = \frac{Q}{\left(\frac{dz}{dx}\right)^2} = \frac{-2 \cos^2 x}{\cos^2 x} = -2$$

$$R_1 = \frac{R}{\left(\frac{dz}{dx}\right)^2} = \frac{2 \cos^4 x}{\cos^2 x} = 2 \cos^2 x = \cancel{2 \cos^2(\sin x)} = 2(1 - \sin^2 x) = 2(1 - z^2)$$

So eqⁿ. becomes,

$$y'' - 2y = 2(1-z^2)$$

The A.E. is,

$$D^2 - 2 = 0$$

$$\Rightarrow D^2 = 2$$

$$\Rightarrow D = \pm \sqrt{2}$$

$$\begin{aligned} \text{C.F.} &= C_1 \cosh \sqrt{2} z + C_2 \sinh \sqrt{2} z \\ &= C_1 \cosh \sqrt{2} \sin x + C_2 \sinh \sqrt{2} \sin x \end{aligned}$$

$$\text{P.I.} = \frac{1}{(D^2 - 2)} \cdot 2(1-z^2)$$

$$= \frac{2}{-2 \left(1 - \frac{D^2}{2} \right)} (1-z^2)$$

$$= - \left[1 - \frac{D^2}{2} \right]^{-1} (1-z^2)$$

$$= - \left(1 + \frac{D^2}{2} + \dots \right) (1-z^2)$$

$$= - (1-z^2) - \frac{D^2}{2} (1-z^2)$$

$$= - (1-z^2) + \frac{2}{2}$$

$$= -1 + z^2 + 1$$

$$= z^2$$

$$= \sin^2 x$$

$$\therefore y = C.F. + P.I.$$

$$= C_1 \cosh \sqrt{2} \sin x + C_2 \sinh \sqrt{2} \sin x + \sin^2 x.$$

~~Transform~~ Reduce to normal form (By changing to dependent variable or by removing first derivative)

Any equation of form,

$$y'' + Py' + Qy = R \quad \text{--- (1)}$$

which is second order differential equation.

Let the solution of the differential (1) be,

$$y = uv \quad \text{--- (2)}$$

where $u = e^{-\int \frac{1}{2} P dx}$

& v is obtained from the reduced normal equation.

$$\frac{d^2 v}{dx^2} + Iv = S \quad \text{--- (3)}$$

where, $I = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx}$

& $S = \frac{R}{u}$

and solving

Putting I & S in eqⁿ. (3), we get the value of v .

Putting u & v in eqⁿ. (2), we get the solution of the given differential equation.

Questions Solve the differential equation:

① $y'' - 2 \tan x y' + y = 0$

Solⁿ:

Comparing the given eqⁿ. with $y'' + Py' + Qy = R$, we get,

$P = -2 \tan x$, $Q = 1$ & $R = 0$

Let the solution,

$y = uv$ — (1)

Now, $u = e^{-\int \frac{1}{2} P dx} = e^{\int \tan x dx} = e^{\log \sec x}$
 $\Rightarrow u = \sec x$

When y is changed to dependent variable then the reduced normal form is,

$$\frac{d^2 v}{dx^2} + I v = S \quad \text{--- (2)}$$

$$\text{Now, } I = 0 - \frac{1}{4} p^2 - \frac{1}{2} \frac{dp}{dx}$$

$$= 1 - \frac{1}{4} (-2 \tan x)^2 - \frac{1}{2} (-2 \sec^2 x)$$

$$= 1 - \frac{1}{4} (4 \tan^2 x) + \sec^2 x$$

$$= 1 - \tan^2 x + \sec^2 x$$

$$= 1 + 1$$

$$= 2$$

$$S = \frac{R}{u} = 0$$

From eqⁿ. (2)

$$\frac{d^2 v}{dx^2} + 2v = 0$$

$$\Rightarrow (D^2 + 2)v = 0$$

The A.E. is,

$$m^2 + 2 = 0$$

$$\Rightarrow m = \pm i\sqrt{2}$$

$$C.F. = C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x$$

~~P.F.~~ ~~(D^2+2)~~

$$\therefore v = C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x$$

From eqⁿ. (1),

$$y = uv$$

$$\therefore y = \sec x (C_1 \cos \sqrt{2}x + C_2 \sin \sqrt{2}x).$$

$$2. \quad y'' + \frac{2}{x} y' - n^2 y = 0$$

Solⁿ:

Comparing the given eqⁿ. with $y'' + Py' + Qy = R$,
we get,

$$P = \frac{2}{x}, \quad Q = -n^2 \quad \& \quad R = 0$$

Let the solution be,

$$y = uv \quad \text{--- (1)}$$

Now, $u = e^{-\int 1/2 P dx} = e^{-\int 1/x dx} = e^{-\log x} = 1/x$

When y is changed to dependent variable then the reduced normal form is,

$$\frac{d^2 v}{dx^2} + Iv = S \quad \text{--- (2)}$$

$$I = 0 - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx}$$

$$= -n^2 - \frac{1}{4} \left(\frac{2}{x} \right)^2 - \frac{1}{2} (-2x^{-2})$$

$$= -n^2 - \frac{1}{x^2} + \frac{1}{x^2}$$

$$= -n^2$$

$$S = \frac{R}{u} = 0$$

From eqⁿ. (2),

$$\frac{d^2 v}{dx^2} - n^2 v = 0$$

$$\Rightarrow (D^2 - n^2)v = 0$$

The A.E. is,

$$D^2 - n^2 = 0$$

$$\Rightarrow D = \pm n$$

$$\therefore C.F. = C_1 e^{nx} + C_2 e^{-nx}$$

$$v = C_1 e^{nx} + C_2 e^{-nx}$$

From eqⁿ. (1),

$$y = \frac{1}{x} C_1 e^{nx} + \frac{1}{x} C_2 e^{-nx}$$

$$3. y'' - 2 \tan x \cdot y' + 5y = \sec x \cdot e^x$$

Solⁿ:

Comparing given eqⁿ. with $y'' + Py' + Qy = R$, we get,

$$P = -2 \tan x, Q = 5 \text{ \& } R = \sec x \cdot e^x$$

Let the solution be,

$$y = uv \text{ — (1)}$$

$$\text{Now, } u = e^{-\int 1/2 P dx} = e^{\log \sec x} = \sec x$$

When y is changed to dependent variable then the reduced normal form is given as,

$$\frac{d^2 v}{dx^2} + I_v = S \text{ — (2)}$$

$$I = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx}$$

$$= 5 - \frac{1}{4} (-2 \tan x)^2 - \frac{1}{2} (-2 \sec^2 x)$$

$$= 5 - \frac{1}{4} (4 \tan^2 x) + \sec^2 x$$

$$= 5 - \tan^2 x + \sec^2 x$$

$$= 5 + 1$$

$$= 6$$

$$S = \frac{R}{u} = \frac{\sec x \cdot e^x}{\sec x} = e^x$$

From eqⁿ. (2),

$$\frac{d^2 v}{dx^2} + 6v = e^x$$

$$\Rightarrow (D^2 + 6)v = e^x$$

The A.E. is,

$$m^2 + 6 = 0$$

$$\Rightarrow m = \pm \sqrt{6}i$$

$$C.F. = C_1 \cos \sqrt{6} x + C_2 \sin \sqrt{6} x$$

$$P.I. = \frac{1 \cdot e^x}{D^2 + 6} = \frac{e^x}{1 + 6} = \frac{e^x}{7}$$

$$\therefore v = C_1 \cos \sqrt{6}x + C_2 \sin \sqrt{6}x + \frac{e^x}{7}$$

From eqⁿ. (1),

$$y = \sec x \left(C_1 \cos \sqrt{6}x + C_2 \sin \sqrt{6}x + \frac{e^x}{7} \right)$$

4. $y'' - 2 \tan x y' - (a^2 + 1)y = e^x \sec x$

Solⁿ:

Comparing given eqⁿ. with $y'' + Py' + Qy = R$, we get,

$$P = -2 \tan x, Q = -(a^2 + 1) \text{ \& } R = e^x \sec x$$

Let the solution be,

$$y = uv \text{ — (1)}$$

$$\text{Now, } u = e^{-\int \frac{1}{2} P dx} = e^{\int \tan x dx} = e^{\log \sec x} = \sec x$$

When y is changed to dependent variable,

$$\frac{d^2 v}{dx^2} + Iv = S \text{ — (2)}$$

$$I = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx}$$

$$= -a^2 - 1 - \frac{1}{4} (-2 \tan x)^2 - \frac{1}{2} (-2 \sec^2 x)$$

$$= -a^2 - 1 - \tan^2 x + \sec^2 x$$

$$= -a^2 - 1 + 1$$

$$= -a^2$$

$$S = \frac{R}{u} = \frac{e^x \cdot \sec x}{\sec x} = \sec x$$

~~Eqn~~ $\therefore$ Eqn (2) becomes,

$$\frac{d^2 v}{dx^2} - a^2 v = \sec x$$

$$\Rightarrow (D^2 - a^2) v = \sec x$$

The A.E. is,

$$D^2 - a^2 = 0$$

$$\Rightarrow D = \pm a$$

$$C.F. = \text{~~Eqn~~} C_1 e^{ax} + C_2 e^{-ax}$$

$$P.I. = \frac{1}{(D^2 - a^2)} \cdot \sec x$$

$$= \frac{1}{-1^2 - a^2} \cdot \sec x$$

$$= \frac{-\sec x}{(a^2 + 1)}$$

$$\therefore v = C_1 e^{ax} + C_2 e^{-ax} - \frac{\sec x}{(a^2 + 1)}$$

$$\therefore y = \sec x \left[C_1 e^{ax} + C_2 e^{-ax} - \frac{\sec x}{(a^2 + 1)} \right]$$

$$5. y'' - 4xy' + (4x^2 - 1)y = -3e^{x^2} \sin 2x$$

Solⁿ:

Comparing the given eqⁿ with $y'' + Py' + Qy = R$, we get,

$$P = -4x, Q = (4x^2 - 1) \text{ \& } R = -3e^{x^2} \sin 2x$$

Let the solution be,

$$y = uv \text{ ————— (1)}$$

$$\text{Now, } u = e^{-\int 1/2 P dx} = e^{2\int x dx} = e^{x^2}$$

When y is changed to dependent variable by removing the first derivative, we get,

$$\frac{d^2 v}{dx^2} + Iv = S \text{ ————— (2)}$$

$$I = Q - \frac{1}{4} P^2 - \frac{1}{2} \frac{dP}{dx}$$

$$= 4x^2 - 1 - \frac{1}{4} (-4x)^2 - \frac{1}{2} (-4)$$

$$= 4x^2 - 1 - 4x^2 + 2$$

$$= 1$$

$$S = \frac{R}{u} = \frac{-3e^{x^2} \sin 2x}{e^{x^2}} = -3 \sin 2x$$

Eqⁿ. (2) becomes,

$$\frac{d^2 v}{dx^2} + v = -3 \sin 2x$$

$$\Rightarrow (D^2 + 1)v = -3 \sin 2x$$

The A.E. is,

$$m^2 + 1 = 0$$

$$\Rightarrow m^2 = -1$$

$$\Rightarrow m = \pm i$$

$$C.F. = C_1 \cos x + C_2 \sin x$$

$$P.I. = \frac{1}{(D^2 + 1)} \cdot (-3 \sin 2x)$$

$$= -3 \cdot \frac{1}{-2^2 + 1} \cdot \sin 2x$$

$$= \sin 2x$$

$$\therefore v = C_1 \cos x + C_2 \sin x + \sin 2x$$

$\therefore$ Eqⁿ. (1) becomes,

$$y = e^{x^2} (C_1 \cos x + C_2 \sin x + \sin 2x)$$